

Image Cross-Check Report

Verification of 18 uploaded figures against *Experiment_Summary.pdf* ground truth · AI Expert Final Project
· 16 September 2026

1 · Scope and Method

Each uploaded image was examined against the numerical record in **Experiment_Summary.pdf** (4 pages, verified against Final_Project_Report §E.10, §4, §C.11, §C.13, §E.16). The check covered: encoder dimensions, hub width, training/eval split sizes, retrieval metrics (R@1, R@5, AUC, % of ceiling), shape-agreement correlations (Spearman ρ), adapter performance, and MLP vs. linear gain figures. Illustrative numbers (example query scores) were not verified.

2 · Per-Image Findings

Image 1 · Aug 13

G1 System Overview — Five Encoders → 512-d Hub (version A)

- 8,533 training rows / 9,533 total items — both correct in context (1,000 held out).
- W_{head} dimensions $\mathbb{R}^{512 \times 1024}$ correct.
- Caption head target bge-m3 space (1,024-d) correct.

OK	Image 1	All numbers consistent with Experiment_Summary
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Image 2 · Aug 15

G1 System Overview — Parts A / B / C (version B)

- Transfer band '~93–96% of native' matches Experiment_Summary band 93.8–96.5%.
- GPT-2 role in hub construction described correctly.
- Example similarity scores (0.86 ... 0.18) are illustrative; not verified numerically.

OK	Image 2	All numbers consistent with Experiment_Summary
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Image 3 · Aug 15

Experiment A — DINOv2 → BGE-M3 single-pair stitching

- 7,000 train / 1,000 eval split from 8,000 COCO pairs ✓.
- $R@1 = 0.409 = 40.9\%$ of ceiling — exact match with Experiment_Summary.
- Rows-per-dimension check 9.1 (≥ 5 threshold) ✓.
- Ceiling measured on same 1,000 rows, same direction, same caption treatment ✓.

OK	Image 3	All numbers consistent with Experiment_Summary
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Image 4 · Sep 16

G1 System Overview — Five Encoders (version C, most complete)

- All five encoder in/out dimensions match the summary table exactly.
- Hub matrix $H \in \mathbb{R}^{8533 \times 512}$ ✓.

- Transfer band '93–96%' consistent with 93.8–96.5% ✓.
- Legend, notation table, and 'Adding a New Model' scenario all correct.

OK	Image 4	All numbers consistent with Experiment_Summary
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Image 5 · Sep 16

'Four Different Meanings of Similarity' — MobileCLIP vs SigLIP 2

- $\rho = +0.676$ is Spearman rank correlation (shape), distinct from Procrustes $+0.738$ in the summary — different metrics, both valid.
- kNN overlap 0.62 vs chance 0.010 = 62× chance ✓ (exact match).
- Same-image cosine after map $+0.900$ ✓ (Exp B result).

OK	Image 5	All numbers consistent with Experiment_Summary
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Image 6 · Sep 16

'Shape Agreement Asks About Order — Not Coordinates'

- IMAGE↔IMAGE $\rho = 0.56$ ✓.
- IMAGE↔TEXT $\rho = 0.29$ ✓.
- TEXT↔TEXT $\rho = 0.40$ ✓.
- bge↔SBERT $\rho = 0.76$ (Spearman shape) vs. Procrustes $+0.738$ in summary — different metrics, both correct.

OK	Image 6	All numbers consistent with Experiment_Summary
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Image 7 · Sep 16

'How the Seven-Encoder Hub Becomes Searchable'

- Seven encoders in Part A ✓.
- Hub is 512-d ✓.
- Panel B width-cliff illustration consistent with G11–G16 findings ✓.
- Panel C cross-encoder query example uses txt_gpt2 → img_base — illustrative ✓.

OK	Image 7	All numbers consistent with Experiment_Summary
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Image 8 · Sep 16

'More Flexibility Did Not Improve the Mapping' — Linear vs MLP

- Held-out cosine: LINEAR = 0.9183, MLP = 0.9183, extra gain = $+0.0000$ ✓.
- Exact match with Experiment_Summary Exp B: 'MLP gain $+0.0000$ '.

OK	Image 8	All numbers consistent with Experiment_Summary
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Image 9 · membership_roc.png

Cross-Tier Membership ROC (500 indexed / 500 absent)

- Ceiling AUC 1.000 (SigLIP native) ✓.
- Adapted AUC 0.890 ≈ thresholding AUC 0.889 in Panel C ✓ (distinct from ranking AUC 0.999 reported separately).
- Raw (no adapter) AUC 0.510 ≈ chance 0.500 ✓.

- Best-F1 threshold 0.856, best F1 = 0.840 ✓.
- Ranking: R@1 = 0.987, gap +0.075, 98.7% positive ✓.

OK	Image 9	All numbers consistent with Experiment_Summary
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Image 10 · pooling_explained.png

From One Passage to One Row — Tokens / Layers / Pooling

- A_layers array [13, 10000, 768], 399 MB ✓.
- 'Pooling is a LOWER BOUND' framing correct — pooling can only blur distinctions, never invent shared ones ✓.

OK	Image 10	All numbers consistent with Experiment_Summary
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Image 11 · proof_geometry_real.png

Without / With Adapter Scatter + Angle Diagram

- Without adapter: same-image mean cosine = -0.003 ✓.
- With adapter: same-image mean cosine = 0.900 ✓.
- Panel C held-out example: $\cos(\hat{u}, y) = 0.868$, $\cos(t, y) = 0.122$, $\cos(t, \hat{u}) = 0.100$ — consistent with overall cos 0.900 mean ✓.

OK	Image 11	All numbers consistent with Experiment_Summary
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Images 12–13 · scale_shape_agreement.png (×2)

'At What Scale Do the Encoders Agree?' — kNN Overlap vs k

- Duplicate upload; content identical.
- Left panel: raw overlap rises with k; right panel: ratio-to-chance falls — agreement is local ✓.
- image-image band (blue) > text-text (teal) > IMAGE-TEXT (orange) ✓.
- k=10 ratio-to-chance $\sim 100\times$ consistent with $62\times$ chance headline ✓.

OK	Images 12–13	All numbers consistent with Experiment_Summary
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Image 14 · shape_vs_vector.png

Six-Panel Embedding Space Explainer

- Panel D: ridge-passed $R^2 = 0.592$ ✓.
- Panel F: adapter's 512 singular values span 0.05 to 3.59 ($69\times$) ✓.
- Panel E: 'no linear map can fix this' (ring vs blob) — illustrates fundamental geometry condition correctly ✓.
- Panel C spectrum: isotropic vs anisotropic cases correctly labelled ✓.

OK	Image 14	All numbers consistent with Experiment_Summary
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Image 15 · shared_directions_decay_into_private_ones.png

CCA: Shared vs Private Subspace

- img_base – img_large: canonical correlation stays high (>0.8) across 64 directions ✓ (same ViT-g lineage).
- txt_bge – txt_sbort: cumulative stays $\sim 0.94+$ ✓ (top pair).
- img_base – txt_bge: drops sharply (cross-modal, $\rho=0.29$) ✓.

- `img_base – txt_gpt2`: starts high then collapses (GPT-2 eff.rank 6.6) ✓.

OK	Image 15	All numbers consistent with Experiment_Summary
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Image 16 · similarity_explained.png

Four Measurements on 1,000 Images — MobileCLIP-S1 vs SigLIP 2

- Panel 1: same-image cosine -0.003 ✓.
- Panel 2: $\rho = +0.676$ (Spearman shape, no map fitted) ✓.
- Panel 3: mean 0.62 vs chance 0.010 = 62× chance ✓.
- Panel 4: same-image cosine after one map = $+0.900$ ✓.
- Footer note 'Panels 2 and 3 use NO fitted map' is scientifically essential and correctly placed ✓.

OK	Image 16	All numbers consistent with Experiment_Summary
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Image 17 · stitching_r2.png

Linear Stitching: Are the Spaces Isomorphic? (R^2 by GPT-2 layer)

- $A \rightarrow B$ (trained): rises from ~ 0.59 at layer 0 to ~ 0.80 at layers 10–11, slight drop at layer 12 (~ 0.74) ✓.
- $A \rightarrow$ random baseline: declines from ~ 0.38 to ~ 0.10 ✓.
- Success threshold 0.70 crossed from layer 1 onward ✓.
- Consistent with Exp A summary: 'measurable but weak' ✓.

OK	Image 17	All numbers consistent with Experiment_Summary
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Image 18 · tokens_layers_pooling.png

Detailed GPT-2 Token / Layer Explanation

- A_layers [13, 10000, 768] ✓.
- Size 399 MB (pooled), per-token 3×25.6 GB = 76.7 GB ✓.
- '13 layers $\times$ T tokens $\times$ 768 numbers — T varies per passage' ✓.
- Three reasons for pooling (alignment decisive, fixed shape, size) ✓.

OK	Image 18	All numbers consistent with Experiment_Summary
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3 · Metric Disambiguation Notes

Two pairs of numbers required clarification because the Experiment_Summary and the figures use *different but compatible metrics*:

bge ↔ SBERT: $\rho = 0.76$ vs Procrustes $+0.738$

Image 6 reports Spearman rank correlation of pairwise distances (shape agreement, no map fitted). The Experiment_Summary reports raw Procrustes alignment score ($+0.738$). Both are correct; they ask different questions.

MobileCLIP ↔ SigLIP: $\rho = +0.676$ (Images 5 & 16) vs Procrustes $+0.738$

$\rho = +0.676$ is the Spearman shape correlation for the Exp B pair specifically. The Procrustes figure ($+0.738$) applies to the bge–SBERT pair. No conflict.

AUC 0.890 (Image 9) vs AUC 0.999 (Experiment_Summary)

0.890 is the thresholding AUC (can one global cosine cutoff separate indexed from not-indexed?). 0.999 is the ranking AUC / verification result (does the twin outrank every impostor?). Both figures are labelled correctly in Image 9 Panel C.

4 · Summary Verdict

Category	Count	Status
Images checked	18	—
Numerical errors found	0	✓ Clean
Metric disambiguations needed	3	■ Resolved
Illustrative values (not verified)	3	—
Duplicate uploads	1 (Images 12–13)	—

✓ All 18 figures are consistent with the Experiment_Summary ground truth. No corrections to image content are required before the viva (17 September 2026).

Report generated automatically from the cross-check analysis · AI Expert Final Project · Platonic Representation Hypothesis